

- 2 Fig. 2 shows a skateboarder of mass 55 kg about to descend a curved ramp in a skate park.

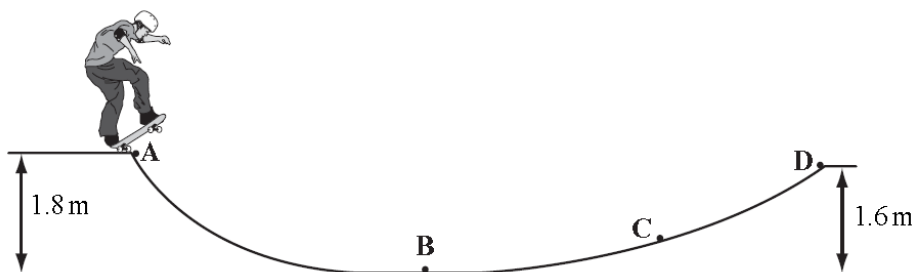


Fig. 2

- (a) At which of the points **A**, **B**, **C** or **D** is the magnitude of the acceleration greatest along the track?

A

..... [1]

- (b) The skateboarder is initially at rest. Assuming that there is no frictional force acting from **A** to **B**, calculate the speed the skateboarder would have at **B**.

statement that $\Delta E_k = \Delta E_p$

$$mgh = \frac{1}{2} mv^2$$

$$v = 5.9(4) \text{ m s}^{-1}$$

kinetic energy = J [2]

- (c) The skateboarder has just enough energy to reach **D** because of friction along **B** to **D**. The length of the track between **B** and **D** is 8.5 m.

Calculate

- (i) the energy lost due to friction as the skateboarder moves from **B** to **D**,

energy lost = $mg \times \text{difference in heights}$

$$= 107.91 \approx 110 \text{ J}$$

energy lost = J [2]

- (ii) the magnitude of the average frictional force.

work done = Fs

$$F = 12.695 \approx 13 \text{ N (ecf from part (c)(i))}$$

average frictional force = N [2]

3 A mass P 80 g is attached to the free end of a horizontal spring on a smooth surface